Performance Testing

Date	04 July 2026
Team ID	SWTID-2026-1614
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing (20 concurrent virtual users, ramped up at 5 users/sec)
Target Module / API	/predict (HDI calculation endpoint) and / (Glassmorphism home page/form)
Test Environment	Cloud / Hosted Server – live production deployment on Render (https://hdi-predictor-ml-flask.onrender.com)
Test Date	4 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Researchers submitting indicators to /predict and receiving simulated HDI calculations.	20	60	All requests succeed; average response time < 2s.
2	Visitors loading the main glassmorphism home page at /.	20	60	Page loads successfully with 0% error rate.
3	Mixed Traffic: Combined GET / and POST /predict traffic to simulate realistic user sessions.	20	60	System remains stable with 0% error rate under mixed load.
4	Idle Recovery: Repeating request execution after idle period to evaluate serverless/free-tier cold-start latency.	20	60	First requests may experience cold start delays; subsequent queries execute quickly.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.85 s (/predict); 1.15 s (/)	Pass	Average latency is well within targets for both serving endpoints.
2	Response Time (Max)	< 5 seconds	12.42 s (/predict); 14.88 s (/)	Fail	Maximum spikes are consistent with Render free-tier container spinning/cold starts, not steady-state calculation issues.
3	Throughput (Req/sec)		2.95 req/s (/predict); 6.12 req/s (aggregated)	N/A	No target limit was set; reported for reference baseline.
4	Error Rate	< 1%	0% (0 failures out of 177 requests to /predict)	Pass	Zero failed requests occurred under load testing limits.
5	CPU Utilization	< 80%	Not measured	N/A	Free hosting tiers do not expose container CPU metrics on developer panels.
6	Memory Utilization	< 80%	Not measured	N/A	Not available on basic dashboards; would require custom profiling agents.

Step 4: Observations & Analysis

Key Findings:
Under a sustained load of 20 concurrent virtual users over 60 seconds, the HDI Predictor's /predict endpoint successfully processed 177 requests with a 0% error rate. The average response time was 850ms, comfortably meeting the <2s target. However, maximum response times peaked at 12.42s for /predict due to initial server wake-up cycles.

Bottlenecks Identified:
The variance between median response times (~320ms) and maximum times (~12s) is characteristic of web hosts spinning down inactive free containers. When the application has not been queried for some time, the container sleeps, causing a cold start while it reinitializes the Python Flask runtime and loads the serialized regression weights (hdi_model.pkl) into memory.

Optimization Steps Taken:
No active speed optimizations have been applied. Future modifications will include implementing periodic cron pings to prevent the web container from entering sleep state, reducing import statements to decrease launch latency, or upgrading to a paid always-on tier if consistent sub-second latency is critical.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Type	Name	Request	Failure C	Median R	Average	Min Resp	Max Resp	Average C	Requests/	Failures/s	50%	66%	75%	80%	90%	95%	98%	99%	99.90%	99.99%	100%
GET	/	180	0	320	1150.4	245.1	14880.3	3210	3.174	0	410	620	980	1250	3200	9800	12500	13800	14500	14700	14880
POST	/predict	177	0	310	850.9	268.6	12420.7	2955.4	2.952	0	400	600	940	1180	2900	9200	10500	11200	12000	12300	12420
	Aggregate	357	0	315	1002.6	245.1	14880.3	3082.9	6.126	0	405	610	960	1220	3050	9500	11200	12800	14200	14600	14880